UNIVERSITY
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Robotics Group Project

Automatic Assembly of a Mobile Robot

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Robotics and Flexible Automation

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Group 2

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1. Introduction

The project involves the design and simulation of a robotic arm (manipulator) to assemble the mobile robot displayed in Figure 1.

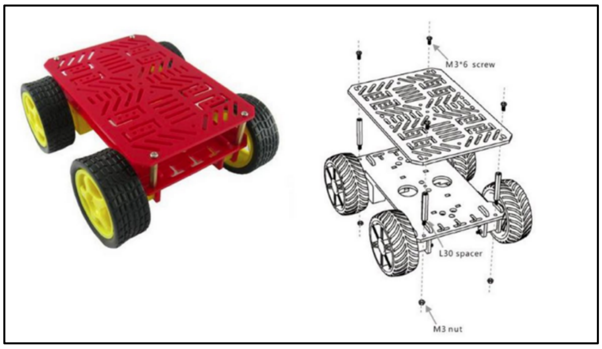

Figure 1 - Assembly process of the mobile robot

The task is to be completed in the assembly cell provided in Figure 2 (not true to scale).

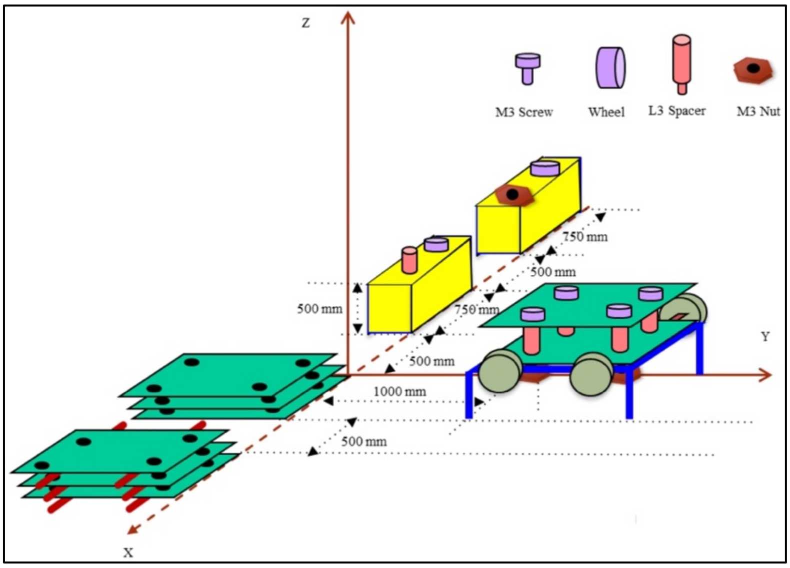

Figure 2 - Assembly cell

The assembly process involves moving between set points 36 different times to assemble each component in the following order:

Figure 3 - Flowchart of assembly process

One of the four corners of the assembly cell is chosen as the base coordinate of the arm to avoid the links passing through each other (making the design of the joints more complicated in a real scenario) when accessing certain locations. Therefore, the base position of the robotic arm with respect to the global coordinate frame is set as (-2750, 1500, 0). The top view of the assembly cell, indicating the position of the robot arm, is displayed in Figure 4.

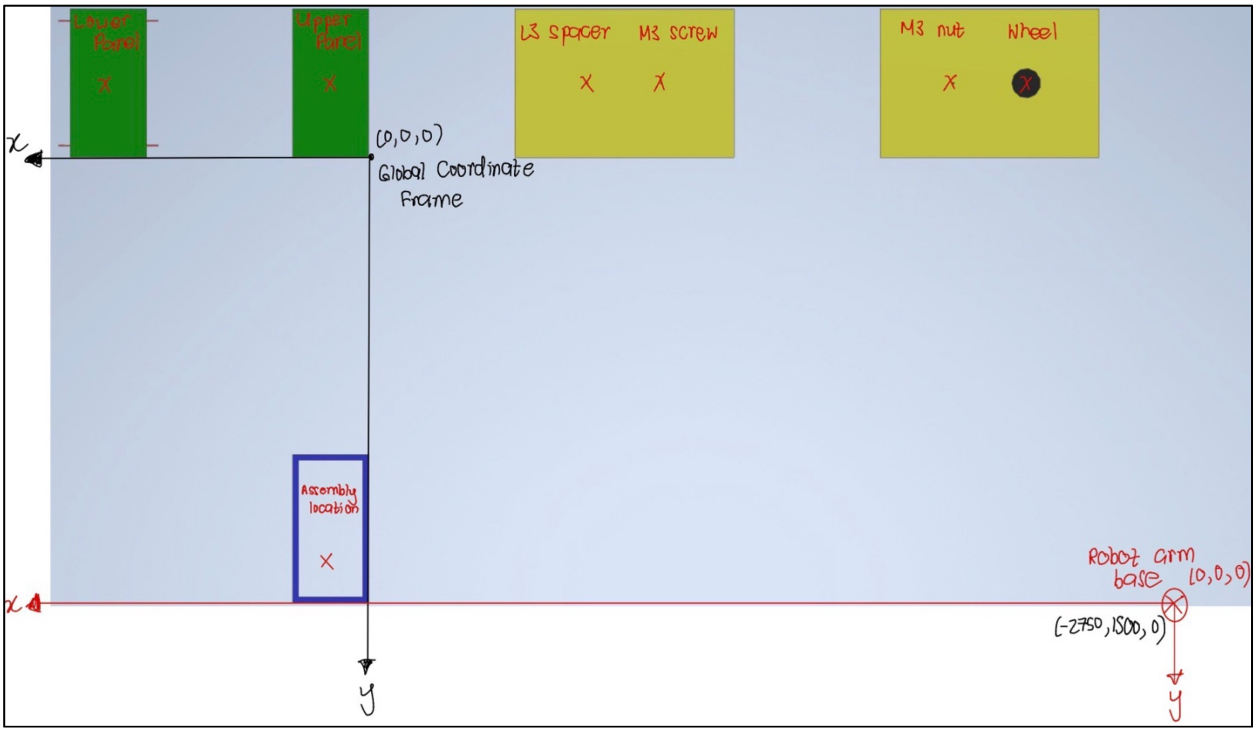

Figure 4 - Top view of assembly cell

The Cartesian coordinates of **each component with respect to the base of the robotic arm** are listed in Table 1.

Table 1 – Cartesian coordinates of components with reference to the base of the robot arm

	Coordinates in mm (x,y,z)
Lower panel	(3640, -1750, 5)
Upper panel	(2880, -1750, 5)
L3 spacers	(2000, -1750, 515)
M3 screws	(1750, -1750, 502)
M3 nuts	(750, -1750, 502)
Wheels	(500, -1750, 515)
Centre-point of assembly location	(2880, -250, 0)

1.1. Project Controls

The task provided the following information to assist in the design of the manipulator:

- Upper and lower panels are 260 x 500 mm.
- Motors that drive the wheels are already installed on the lower panel.
- Wheels have a push-to-clamp assembly with more shafts
- L3 spacer is 30 mm.

1.2. Project Assumptions

The assumptions below were considered as part of the assessment:

- The design must consist of only one robot. Fixtures may be used to assist the robot.
- Gripping mechanism is not considered; it is assumed the end effector is always equipped with the right gripper.
- Each assembly component is approached from above, normal to its plane.
- Dimensions that are not already provided may be assumed.
- Fingers of the robot gripper (tool) have a length of 40 mm.

Additional assumptions were made to justify the practicality of the design. They are listed below:

- The stability and mechanical aspects of the robot are ignored.
- The L3 spacers, M3 screws, M3 nuts and wheels are available from a spring-activated parts feeder at the same locations.
- The robot arm base coordinate frame is parallel to the global coordinate frame.
- The M3 screw has right-hand threads (tightens when turned clockwise).

Table 2 - Assumed dimensions of components

Feature	Dimension
Height of lower panel centre point	5 mm
Distance to centre of screw holes on lower and upper panels	20 mm from edge of lower panel
Distance to motor shafts on lower panel	40 mm from edge of lower panel
Height raised by assembly table (blue)	100 mm
Diameter of motor shaft	5 mm
Height of wheel	30 mm
Centre-point of screw, wheel, spacer and nut from the corner of holding block (yellow)	250 mm

2. Section A: Designing the Robot

2.1. Characteristics of the robotic arm

The robot is designed to make it as simple as possible to satisfy the task. The table below highlights the characteristics of the robot design:

Table 3 - Characteristics of robot arm design

<table>
<tr><td>Degrees of freedom (DOF)</td><td colspan="2">5</td></tr>
<tr><td>Number of Joints</td><td colspan="2">5 Joints: PPPRR</td></tr>
<tr><td rowspan="4">Length of links</td><td>Link 1</td><td>630 mm</td></tr>
<tr><td>Link 2</td><td>3640 mm</td></tr>
<tr><td>Link 3</td><td>1750 mm</td></tr>
<tr><td>Link 4</td><td>60 mm</td></tr>
<tr><td>Tool offset (fingers of gripper)</td><td colspan="2">40 mm</td></tr>
</table>

Although a 5-DOF robotic arm may be considered deficient for more complex tasks, it is fully capable of performing the required assembly. To keep the analysis as straightforward as possible, the design uses the minimum number of degrees of freedom, replacing as many revolute joints as possible with simpler prismatic joints.

The three prismatic joints provide the end-tool full access to any location within the assembly cell, while the two revolute joints perform special functions. One revolute joint enables the screwing of M3 fasteners, and the other allows the wheels to be reoriented from their horizontal orientation during pick-up to a vertical orientation for clamping onto the lower panel.

The link lengths were determined such that the tool can reach all component pick-up positions as well as their corresponding assembly points. While these lengths may vary considerably, the design objective of the robot is to complete the assembly with minimal functional requirements.

Horizontal and vertical reaches

The workspace is not symmetric as there is no revolute joint at the base of the manipulator. The maximum reach in the horizontal (X-Y) plane is listed below:

- *3640 mm in the x-axis*
- *1750 mm in the y-axis*

With the help of the revolute joint at the wrist, a further 80 mm is offset (addition of the tool fingers). Therefore, the furthest distance the tool tip can reach from the base of the robot is 4111 mm.

The first prismatic joint enables the robot arm to move in the vertical direction, and its movement is limited by only the maximum height of the first link. The stroke within the vertical axis is 20 mm lower as link 4 is offset down by 20 mm to house the first revolute joint. The maximum reach and stroke in the vertical (Z-Y) plane are listed below:

- *Reach: 630 mm in the z-axis*
- *Stroke: 610 mm in the z-axis*

Figure 5 displays the horizontal and vertical reach of the robot highlighted in the CAD model.

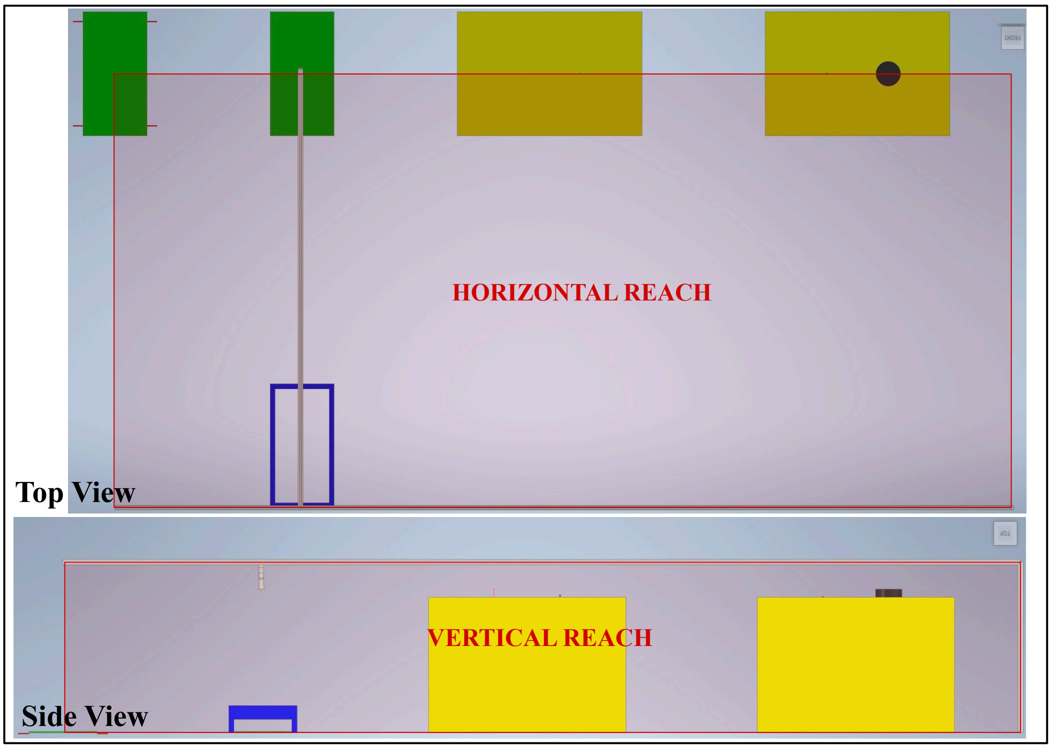

Figure 5 - Horizontal (top view and vertical reach displayed on assembly cell

Joint space work envelope

Limits were assigned to each joint to ensure they are constrained to operate within the assembly cell. Figure 6 shows a visual representation of the joint space work envelope:

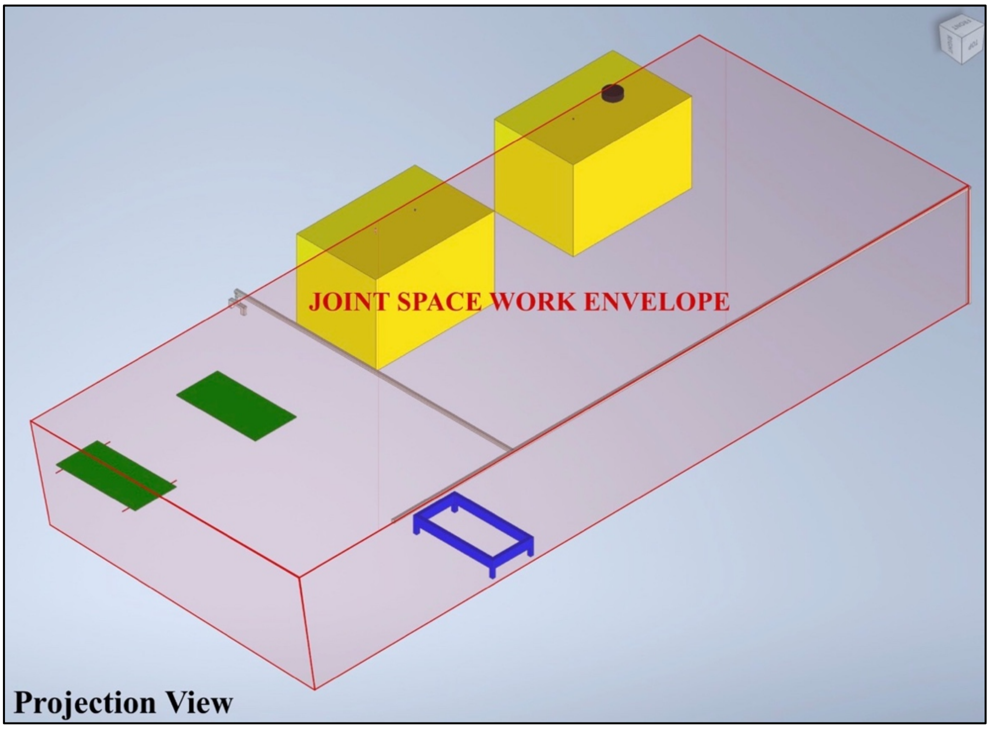

Figure 6 - Joint space work envelope in CAD model

Each 'q' represents a joint with reference to the base of the robot arm. The matrix below represents the minimum and maximum limits for each joint.

$$[q_{min}] \leq [C][q] \leq [q_{max}]$$

$$\begin{bmatrix} 0 \\ 0 \\ -1750 \\ -\frac{\pi}{2} \\ -\pi \end{bmatrix} \leq \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q1 \\ q2 \\ q3 \\ q4 \\ q5 \end{bmatrix} \leq \begin{bmatrix} 630 \\ 3640 \\ 0 \\ \frac{\pi}{2} \\ \pi \end{bmatrix}$$

2.2. Tool orientation and location

The tool orientation and location are determined when picking up a wheel and a lower panel. The orientation is determined with reference to the robot's base coordinate frame. When selecting the tool tip location, the tool centre point (TCP) is positioned at the centre of the component with a **tool offset of 40 mm** along the z-axis (approach) to account for the gripper fingers. The figure below displays the tool when performing each of the tasks:

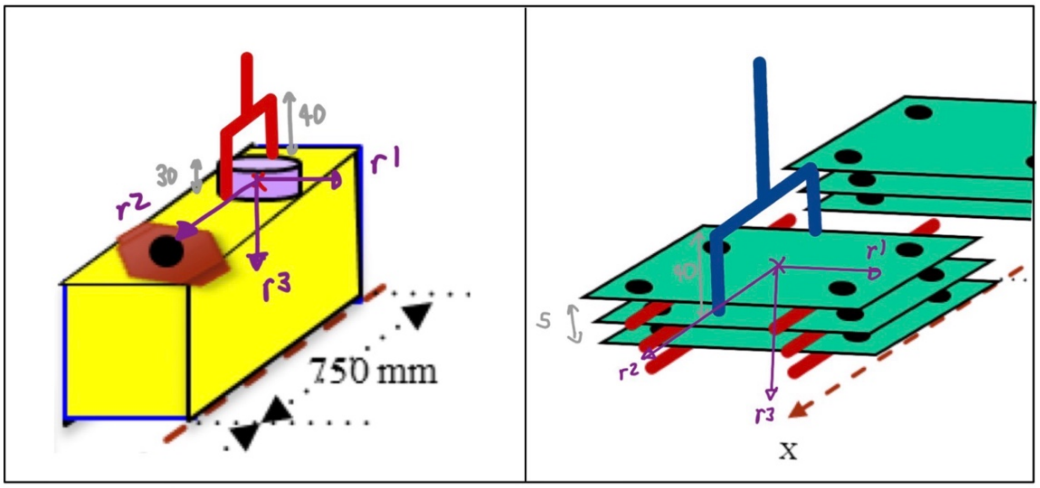

Figure 7 - Tool position and orientation when picking up a wheel (left) and lower panel (right)

1) Picking up a wheel

$$p = \begin{bmatrix} 500 \\ -1750 \\ 500 + \frac{30}{2} + 40 \end{bmatrix}$$

$$R = R_y(180°)R_z(90°) = \begin{bmatrix} \cos(180) & 0 & \sin(180) \\ 0 & 1 & 0 \\ -\sin(180) & 0 & \cos(180) \end{bmatrix} \begin{bmatrix} \cos(90) & -\sin(90) & 0 \\ \sin(90) & \cos(90) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0 & 1 & 0 & 500 \\ 1 & 0 & 0 & -1750 \\ 0 & 0 & -1 & 555 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2) Picking up the lower panel

$$p = \begin{bmatrix} 3640 \\ -1750 \\ 5 + 40 \end{bmatrix}$$

$$R = R_y(180°)R_z(90°) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$T = \begin{bmatrix} 0 & 1 & 0 & 500 \\ 1 & 0 & 0 & -1750 \\ 0 & 0 & -1 & 45 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.3. Kinematic model of the robot

The coordinate frames are attached as per the Denavit-Hartenberg (DH) algorithm.

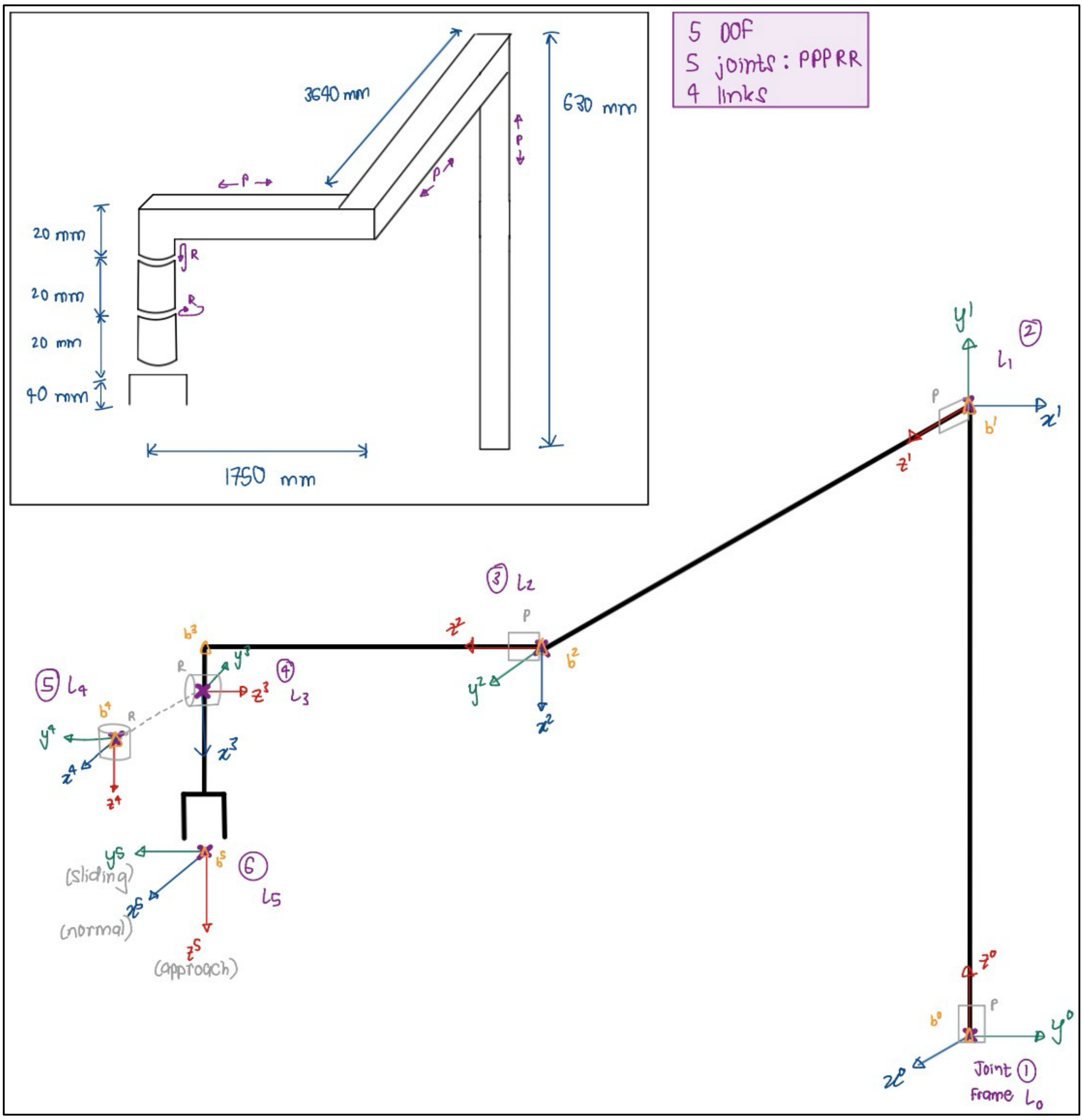

Figure 8 - Robot arm design

Table 4 - Denavit-Hartenberg (DH) parameters

Axis	θ	d	a	α	home
1	90º	q(1)	0 mm	90º	630 mm
2	-90º	q(2)	0 mm	90º	3640 mm
3	0º	q(3)	20 mm	-180º	1750 mm
4	q(4)	0 mm	0 mm	-90º	-90º
5	q(5)	80 mm	0 mm	0º	0º

From the DH parameters, the individual transformation matrix for each joint is obtained as shown below.

$$T_k^{k+1} = \begin{bmatrix} \cos(\theta) & -cos(\alpha)sin(\theta) & \sin(\alpha)\sin(\theta) & a\cos(\theta) \\ sin(\theta) & cos(\alpha)cos(\theta) & -sin(\alpha) & a\sin(\theta) \\ 0 & \sin(\alpha) & cos(\alpha) & d \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$T_0^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & q(1) \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_1^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & q(2) \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_2^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & q(3) \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$T_3^4 = \begin{bmatrix} cos\theta_4 & 0 & sin\theta_4 & 0 \\ sin\theta_4 & - & sin\theta_4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\ T_4^5 = \begin{bmatrix} \cos(\theta_5) & -\sin(\theta_5) & 0 & 0 \\ \sin(\theta_5) & -\cos(\theta_5) & 0 & 0 \\ 0 & 0 & 1 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The overall arm transformation is shown below

$$T_0^5 = \begin{bmatrix} \cos(\theta_4)\cos(\theta_5) & cos(\theta_4)\sin(\theta_5) & sin(\theta_4) & q(1) - 0.1sin(\theta_4) \\ \sin(\theta_5) & \cos(\theta_5) & 0 & q(2) \\ -\sin(\theta_4)\cos(\theta_5) & \sin(\theta_4)\sin(\theta_5) & \cos(\theta_4) & q(3) - 0.1cos(\theta_4) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2.4. Tool characteristics of the robot

Tool configuration vector

$$Tool\ configuration\ vector = \begin{bmatrix} \frac{2}{25}\sin(q(4) - q(2)) \\ q(1) - q(3) - \frac{2}{25}\cos(q(4)) \\ \frac{q(5)}{\pi}\cos(q(4)) \end{bmatrix}$$

Jacobian matrix

$$J = \begin{bmatrix} 0 & 0 & 0 & \frac{2\cos(q(4))}{25} & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & -1 & \frac{2\sin(q(4))}{25} & 0 \\ 0 & 0 & 0 & -\frac{q(5)\cos(q(4))}{\pi} & -\frac{\sin(q(4))}{\pi} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{q(5)\sin(q(4))}{\pi} & \frac{\cos(q(4))}{\pi} \end{bmatrix}$$

3. Section B: ARTE & Simulation

The following tasks are completed using A Robotic Toolbox for Education (ARTE), a MATLAB toolbox that allows simulation and visualisation of manipulators.

3.1. ARTE model of the robot

The design of the robot arm and the assembly cell was modelled in Autodesk Inventor. The parts match the design from section A, and each link was modelled as per their dimensions in their respective position and orientation. The links were then exported in the .stl format and save as *linkx_base.stl* files on the robot arm folder.

The base files were transformed to reference systems in moving frame using the *transform_to_own* function. This function was also used to convert the dimensions from millimetres to metres. The *import_links.m* script was used to import the files.

A parameters file was copied from a similar robot and edited to match the design of the project. Changes were made to the robot's name, robot path, DH table, DOF, type of joints and robot limits to match the design. The teach function was then used to test the model of the robot. Figure below displays the robot being executed (using *run_robot.m* script) upon being imported into ARTE.

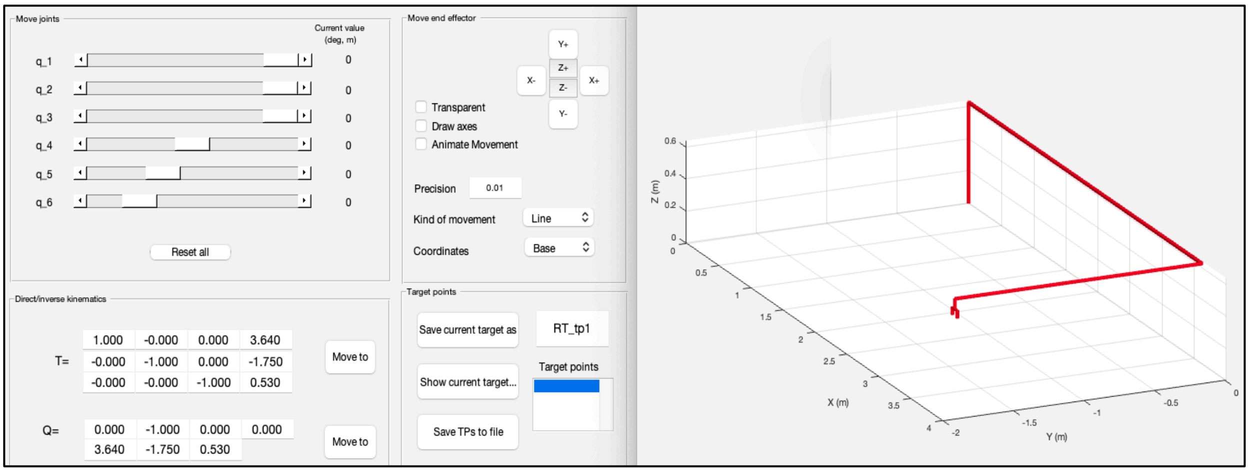

Figure 9 - Robot model imported in ARTE

3.2. Kinematic model of the robot

The kinematic model was systematically derived using ARTE's built-in functions. The Denavit-Hartenberg method was applied to compute the transformation of matrices step by step.

Test Configuration

For the derivation, the joint configuration q = [0.3, 1.0, -.08, 0, 0] was used. This was because it represents a meaningful position within the robot's operational workspace.

Denavit-Hartenberg Parameters

The Denavit-Hartenberg parameters extracted from ARTE are presented in Table 5.

Table 5 - Denavit-Hartenberg (DH) parameters from ARTE

Joint	θ	d	a	α
1	$\frac{\pi}{2}$	$q_1 + 0.63$	0	$\frac{\pi}{2}$
2	$-\frac{\pi}{2}$	$q_2 + 3.64$	0	$\frac{\pi}{2}$
3	0	$q_3 + 1.75$	0.002	$-\pi$
4	$q_4 - \frac{\pi}{2}$	0	0	$-\frac{\pi}{2}$
5	q_5	0.08	0	0

Individual Transformation Matrices
Using ARTE's *dh()* function, each transformation matrix was calculated:

T_0^1 (Base to Joint 1)

```
T01 (Base to Joint 1):
    0.0000   -0.0000    1.0000         0
    1.0000    0.0000   -0.0000         0
         0    1.0000    0.0000    0.9300
         0         0         0    1.0000
```

T_1^2 (Joint 1 to Joint 2)

```
T12 (Joint 1 to Joint 2):
    0.0000    0.0000   -1.0000         0
   -1.0000    0.0000   -0.0000         0
         0    1.0000    0.0000    4.6400
         0         0         0    1.0000
```

T_2^3 (Joint 2 to Joint 3)

```
T23 (Joint 2 to Joint 3):
    1.0000         0         0    0.0200
         0   -1.0000    0.0000         0
         0   -0.0000   -1.0000    0.9500
         0         0         0    1.0000
```

T_3^4 (Joint 3 to Joint 4)

```
T23 (Joint 2 to Joint 3):
    1.0000         0         0    0.0200
         0   -1.0000    0.0000         0
         0   -0.0000   -1.0000    0.9500
         0         0         0    1.0000
```

T_4^5 (Joint 4 to Tool)

```
T45 (Joint 4 to Joint 5):
    1.0000         0         0         0
         0    1.0000         0         0
         0         0    1.0000    0.0800
         0         0         0    1.0000
```

Composite Transformation
The complete transformation from base to tool was computed as:
T_0^5 (Base to Tool)

```
Final Transformation Matrix T05:
    1.0000   -0.0000    0.0000    4.6400
   -0.0000   -1.0000    0.0000   -0.9500
   -0.0000   -0.0000   -1.0000    0.8300
         0         0         0    1.0000
```

Comparison with Manual Calculations
The derivation done by ARTE and the manual calculations represent complementary approaches to kinematic modelling:

- The ARTE approach was numerical computation using built-in functions with standard DH conventions
- The manual approach was an analytical derivation providing symbolic understanding of kinematic relationships.

The kinematic model was successfully derived using ARTE, producing numerically accurate transformation matrices. While the numerical results are not the same, both approaches are valid and complement one another. The manual approach provides symbolic equations that deepen theoretical insight, whereas the ARTE approach delivers numerical data ready for simulation. Together, they offer a well-rounded understanding of the robot's kinematics, bridging theory and practical application.

3.3. Tool Jacobian matrix

The tool-configuration Jacobian matrices were determined for the M3 screw assembly operations using ARTE's manipulator Jacobian function. Two critical configurations were analysed during the screw assembly process.

Initial ARTE Implementation Results

- Jacobian when picking up M3 Screw (q = [0.562. 1.750, -1.750, 0, 0])

```
Jacobian Matrix (6x5) when PICKING M3 screw:
         0    1.0000         0   -0.0800   -0.0000
         0   -0.0000   -1.0000    0.0000    0.0000
    1.0000    0.0000   -0.0000    0.0000    0.0000
         0         0         0    0.0000    0.0000
         0         0         0    1.0000    0.0000
         0         0         0    0.0000   -1.0000
```

- Jacobian when depositing M3 Screw (q = [0.562. 2.2770, -0.250, 0, 0])

```
Jacobian Matrix (6x5) when DEPOSITING M3 screw:
         0    1.0000         0   -0.0800   -0.0000
         0   -0.0000   -1.0000    0.0000    0.0000
    1.0000    0.0000   -0.0000         0         0
         0         0         0    0.0000    0.0000
         0         0         0    1.0000    0.0000
         0         0         0    0.0000   -1.0000
```

Singularity Analysis
Both Jacobian matrices demonstrated full rank (3/3) in their position components, indicating that the robot operates away from singular configurations during screw assembly operations. This confirms that the robotic arm maintains full mobility and can execute the required pick-and-place tasks without encountering kinematic singularities that would limit its dexterity.

Physical Interpretation
- Linear motion in the X-direction is primarily controlled by joint 1 (prismatic)
- Y-direction motion is controlled by joint 2 (prismatic)
- Z-direction motion is controlled by joint 3 (prismatic)
- The revolute joints (4 and 5) provide orientation control with coupling to position through tool offset.

NOTE: Corrected implementation for manual calculation alignment.
To ensure consistency with the manual calculations in Section A, the DH parameters were modified by removing constant offsets from the joint variables. The corrected implementation yielded the following Jacobian matrices:

The amended code in the *parameters.m* file is as below:

```
% DH table (in meters and rad)(use these parameters to obtain the same Jacobian
matrix as Section A manual calculation)
robot.DH.theta = '[pi/2 -pi/2 0 q(4) q(5)]'; % Remove -pi/2 offset
robot.DH.d = '[q(1) q(2) q(3) 0 0.08]'; % Remove constant offsets
robot.DH.a = '[0 0 0.02 0 0 ]'; % Keep 0.02m for joint 3
robot.DH.alpha = '[pi/2 pi/2 -pi -pi/2 0 ]'; % No change required
```

- New Jacobian when picking the M3 screw

```
Jacobian Matrix (6x5) when PICKING M3 screw:
         0    1.0000         0         0         0
         0   -0.0000   -1.0000   -0.0000    0.0000
    1.0000    0.0000   -0.0000    0.0800    0.0000
         0         0         0    0.0000   -1.0000
         0         0         0    1.0000    0.0000
         0         0         0    0.0000   -0.0000
```

- New Jacobian when depositing the M3 screw

```
Jacobian Matrix (6x5) when DEPOSITING M3 screw:
         0    1.0000         0   -0.0000    0.0000
         0   -0.0000   -1.0000   -0.0000    0.0000
    1.0000    0.0000   -0.0000    0.0800   -0.0000
         0         0         0    0.0000   -1.0000
         0         0         0    1.0000    0.0000
         0         0         0    0.0000   -0.0000
```

Comparison with Manual Calculations
The corrected Jacobian matrices show significantly improved alignment with the manual calculations from Section A. The structural similarities confirm consistent kinematic behaviour with both implementations demonstrating:

- Identical linear velocity characteristics for prismatic joints.
- Consistent coupling between revolute joints and tool position.
- Full rank position Jacobians (3/3) confirming non-singular operation.
- Proper tool configuration vector relationships.

The minor differences in specific matrix elements are attributable to different mathematical formulations and tool configuration definitions between the manual analytical approach and ARTE's numerical implementation.

Both implementations validate that the robotic arm design provides adequate dexterity and mobility for the screw assembly task, operating away from kinematic singularities throughout the required workspace. The consistency between the manual and ARTE analyses confirms the robustness of the kinematic model.

3.4. Tooltip position and orientation of the robot

We aim to determine the tool tip position and orientation for four representative joint configurations using the ARTE toolbox in MATLAB. These joint vectors are chosen based on the component coordinates provided in the assembly cell and what we have calculated previously. The four chosen configurations correspond to key stages of the assembly process: wheel pickup, lower panel pickup, upper panel pickup, and the assembly centre.

A joint vector can be written as:

$$q = [q1, q2, q3, q4, q5]$$

where q1, q2, q3 are prismatic joints (in metres) and q4, q5 are revolute joints (in radians).

q1 represents the vertical prismatic joint (z-axis lift), q2 represents the horizontal prismatic joint (x-axis extension), and q3 represents the horizontal prismatic joint (y-axis translation). While q4 represents the revolute wrist joint (tilt/pitch) and q5 represents the revolute wrist joint (roll/rotation about the tool axis).

The four configurations with the corresponding joint vectors that we are going to study are the following:

1. Wheel pick-up

$$q = [0.555, 0.500, -1.750, 0, 0]$$

2. Lower panel pickup

$$q = [0.045, 3.640, -1.750, 0, 0]$$

3. Upper panel pickup

$$q = [0.045, 2.880, -1.750, 0, 0]$$

4. Assembly centre

$$q = [0.040, 2.880, -0.250, 0, 1.571]$$

The MATLAB Code is deconstructed to help find the tool tip position and orientation, as well as the configuration for each configuration.

The robot model (5-DOF, PPPRR) is loaded into ARTE using the *load_robot* function upon initialisation with *init_lib*.

The four q-vectors are computed on MATLAB:

```
QS = [
    0.555, 0.500, -1.750, 0,     0;
    0.045, 3.640, -1.750, 0,     0;
    0.045, 2.880, -1.750, 0,     0;
    0.040, 2.880, -0.250, 0, 1.571
];
```

For each configuration, we evaluate the forward kinematics using *T = directkinematic(robot,q)*. This computes the homogeneous transform T (4x4). We then extract the position and orientation for each configuration using:

```
pos = T(1:3,4);
R = T(1:3,1:3);
eul = rotm2eul(R,'ZYX');
rpy = rad2deg(eul([3 2 1]));
```

Finally, we obtain the following tool tip position and orientation for the four representative joint configurations:

<u>Configuration 1</u>
Joint vector q = [0.555, 0.500, -1.750, 0.000, 0.000]
TCP Position (x,y,z) [m] = [4.140, -0.000, 1.085]
Orientation (roll, pitch, yaw) [deg] = [-180.0, 0.0, -0.0]

<u>Configuration 2</u>
Joint vector q = [0.045, 3.640, -1.750, 0.000, 0.000]
TCP Position (x,y,z) [m] = [7.280, -0.000, 0.575]
Orientation (roll, pitch, yaw) [deg] = [-180.0, 0.0, -0.0]

Configuration 3
Joint vector q = [0.045, 2.880, -1.750, 0.000, 0.000]
TCP Position (x,y,z) [m] = [6.520, -0.000, 0.575]
Orientation (roll, pitch, yaw) [deg] = [-180.0, 0.0, -0.0]

Configuration 4
Joint vector q = [0.040, 2.880, -0.250, 0.000, 1.571]
TCP Position (x,y,z) [m] = [6.520, -1.500, 0.570]
Orientation (roll, pitch, yaw) [deg] = [180.0, 0.0, -90.0]

Furthermore, we want to visualise the configurations using MATLAB in the assembly workspace. To do so, we create a 3D figure and set up a plot.

```
figure; hold on; grid on;
xlabel('X (m)'); ylabel('Y (m)'); zlabel('Z (m)');
title('TCP positions for 4 configurations');
axis equal; view(120,30);
```

Each configuration is then plotted using the robot's joint values, with the forward kinematics function computing the position and orientation.

```
for i = 1:4
q = QS(i,:);
T = directkinematic(robot,q);
pos = T(1:3,4);
```

We add the tool frame for each configuration and label them:

```
plot3(pos(1),pos(2),pos(3),'o','MarkerSize',8,'MarkerFaceColor','r');

draw_axes(T,'X','Y','Z',0.6);
text(pos(1),pos(2),pos(3)+0.05,sprintf('Config %d',i),'FontSize',10);
```

Finally, we graph the result as seen in Figure 10.

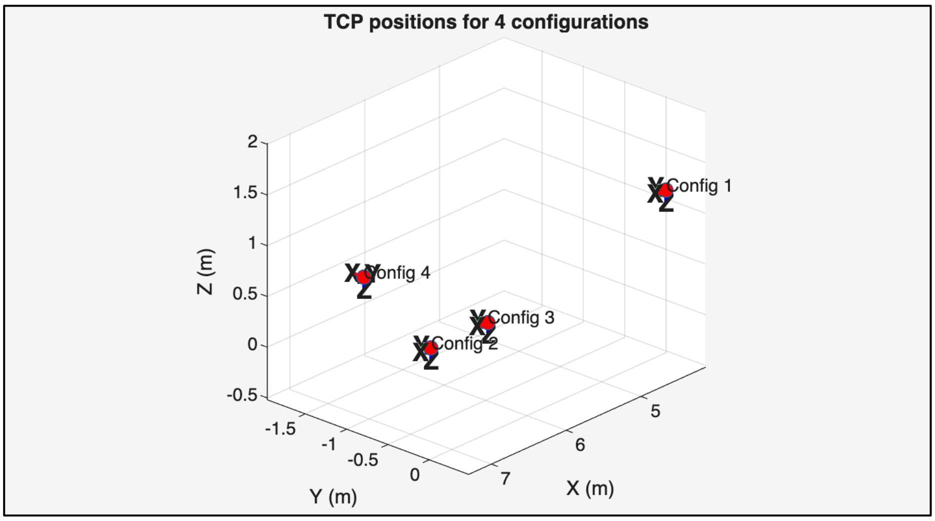

Figure 10 - Tool configuration positions

The visualisation of the four TCP frames in the assembly workspace shows that the robot can reach all critical task positions: wheel pickup, lower panel pickup, upper panel pickup, and the assembly centre.

3.5. Smooth transition for the tooltip

As the robot moves an M3 screw from its feeder position to the first hole on the panel below on the left, the goal is to provide a smooth and continuous trajectory for the tooltip. To ensure that the tool orientation remains normal with respect to the panel surface, the transition must preserve collision-free mobility. ARTE (A Robotic Toolbox for Education) is used in MATLAB to run the simulation, allowing workspace trajectories and joint movements to be visualized. The figure illustrates a smooth 3D trajectory to connect the Nut Feeder, mid-air via point (TCP), and Bolt locations.

The red line shows the robot's tooltip path by using the pick-and-place motion, to ensure a collision-free as well as continuous transition. Coordinate frame (Z(feeder), Z(TCP), Z(bolt)) is used to indicate orientation at key stages of the movement within the workspace.

Table 6 - Cartesian waypoints defining the up-across-down trajectory

Point	Description	(x,y,z) in meters	Notes
P	Screw pick-up	(1.750, -1.750,0.502)	Feeder location
AP	Above-pick	(1.750, -1.750,0.562)	+60 mm clearance: 40 mm fingers + margin
AL	Above-Hole	(2.770, -0.250,0.562)	Same Z as AP for flat traverse
L	First left hole	(2.770, -0.250,0.502)	Target hole for Screw placement

Table 7 – Joint configurations used to defin start and end positions

Pose	Joint vector q = [q1 q2 q3 q4 q5]
Pick-up (P)	[0.555, 0.500, -1.750 ,0 ,0]
Hole (L)	[0.040, 2.880, -0.250, 0, 1.571]

Intermediate via-points were generated by adding +0.060 m to q1 at P (AP) and by matching q2-q3 of L while keeping the clearance q1(AL)

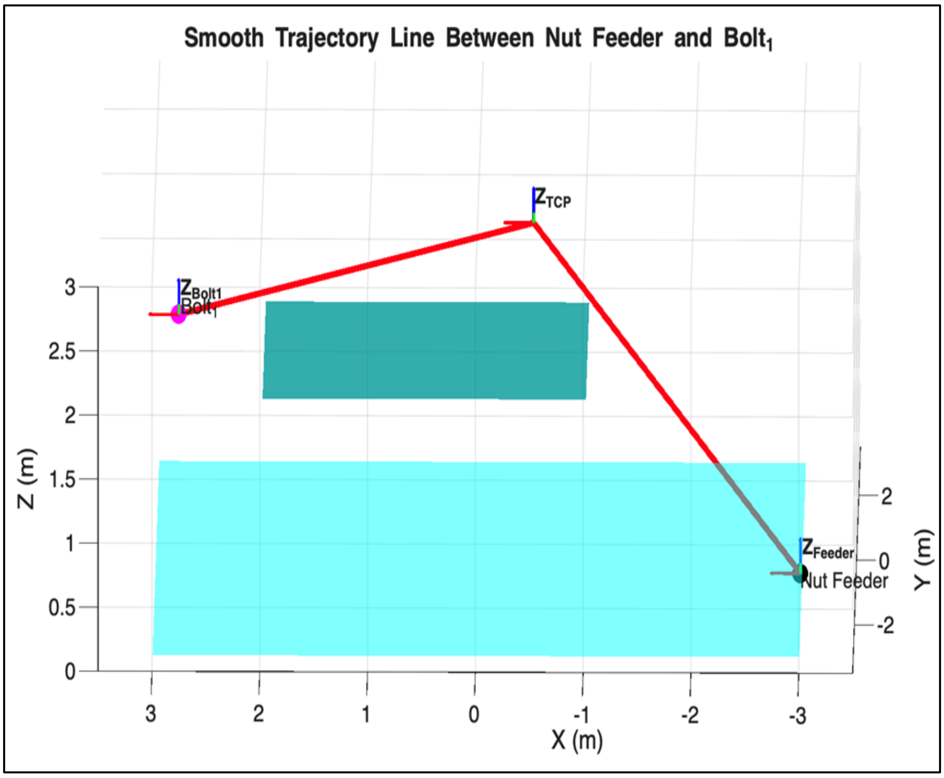

Figure 11 – Smooth trajectory line between the nut feeder and bolt

3.6. Tool trajectory and joint velocities

In robotic motion planning, the tool trajectory defines the precise path followed by the Tool Centre Point (TCP) as the manipulator moves between target positions such as pick and place locations. It represents the spatial movement of the robot's end-effector in Cartesian coordinates (X, Y, Z), ensuring smooth, collision-free, and efficient task execution.

Along this trajectory, the joint velocities describe how fast each joint of the robot must move to achieve the desired end-effector motion. Accurate calculation of joint velocities is essential for maintaining smooth motion, minimizing dynamic stresses, and ensuring coordinated movement across all joints.

Together, tool trajectory generation and joint velocity analysis form the foundation of robotic path planning, enabling precise, safe, and efficient automation in industrial applications.

Calculation of joint velocities:
In a Cartesian manipulator (with three prismatic joints), the relationship between joint variables and end-effector position is linear. Since there are no rotational joints, the joint velocities are directly equal to the Cartesian velocities of the end-effector

Let's start from the assembly data
Assembly centre point (where panel is placed):
(x,y,z) = (2.880, -0.250, 0.00)

Panel size: 0.260 m × 0.500 m (X × Y).
Hole offset: 0.020 m from the left edge along X.

These define the robot's workspace points and safe approach distances.

Now we will compute the first hole position (L)
X-coordinate: assembly centre (2.880) – half panel length (0.260/2 = 0.130) + hole offset (0.020)

X = 2.880 – 0.130 + 0.020 = 2.770.
This locates the hole position (L) relative to the panel's centre.
This step ensures correct placement alignment before vertical motion.
The gripper fingers are 40 mm long, and we should add a safe clearance distance of 60 mm (finger length + margin) when approaching/retreating vertically.

So, we add that clearance to the screw's Z height:
z{AP} = 0.502 + 0.060 = 0.562
This defines the Above-Pick (AP) point — a safe vertical offset above the pick surface.

Now we'll break it down the joint velocities step-by-step for our three prismatic joints (q1, q2, q3)

Now, from Forward Kinematics
With (orientation fixed):
q1 = x, q2 = y, q3 = z
So, each prismatic joint velocity is just the corresponding Cartesian velocity.

Now segment-by-segment calculation
(a) Above-pick → Pick (AP → P)
Motion: straight down in Z by 0.060 m in 1 s.
Joint velocities:
q1 = 0, q2 = 0, q3 = -0.060
Only the Z-joint moves; hence, only is nonzero.
Negative sign shows downward motion

(b) Pick → Above-pick (P → AP)
Motion: straight up in Z by 0.060 m in 1 s.
Joint velocities:
q1 = 0, q2 = 0, q3 = +0.060
Only the Z-joint moves; hence, only is nonzero.
Reverse of previous motion, same magnitude, opposite direction.

(c) Above-pick → Above-place (AP → AL)
This is a 3D diagonal motion; all three prismatic joints move simultaneously to translate the gripper from pick to place region while keeping orientation fixed. Velocity components indicate the relative contribution of each axis to the trajectory.

Δ x = 2.770– 1.750 = 1.020
Δy = -0.250 – (-1.750) = -0.250 + 1.750 = 1.500
Δz = 0.165 – 0.562 = -0.397

Motion: displacement vector.
(Δx, Δy, Δz) = (1.020, 1.500-0.397)
Distance = 1.793m
Duration chosen = 1.793/0.2 ~= 8.97
 x = {1.020}/{8.97} = 0.1137
 y = {1.500}/{8.97} = 0.1673
 z = {-0.397}/{8.97} = -0.0443
q1 = 0.1137, q2 = 0.1673, q3 = -0.0443

(d) Above-place → Place (AL → L)
Motion: straight down in Z by 0.060 m in 1 s.
Joint velocities:
q1 = 0, q2 = 0, q3 = -0.060

Final descent motion, the gripper lowers the part gently to the placement hole.

(e) Place → Above-place (L → AL)
This is a retraction motion — once the part is placed, the gripper retreats safely upward, avoiding collisions with the assembly surface.

Motion: straight up in Z by 0.060 m in 1 s.
Joint velocities:
q1 = 0, q2 = 0, q3 = +0.060

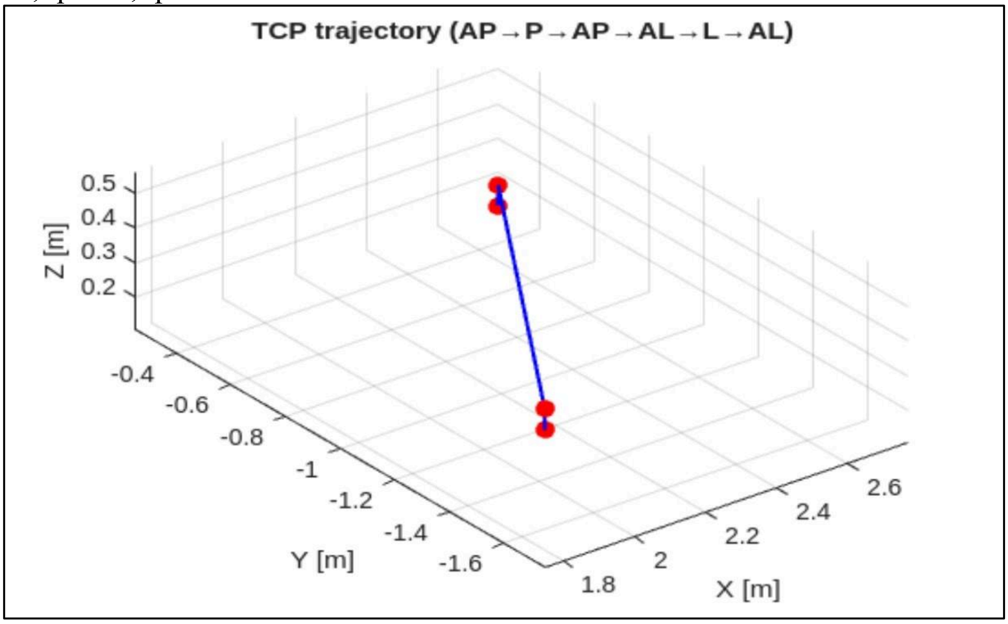

Figure 12 – Tool trajectory for M3 screw from pick-up to place

- **AP (Above Pick)**: Starting position above the object to be picked.
- **P (Pick)**: The point where the tool lowers to pick up the object.
- **AP (Above Pick again)**: The tool retracts upward to a safe height after picking.
- **AL (Above Place)**: The tool moves horizontally to the position above where the object will be placed.
- **L (Place)**: The tool lowers to place the object. A
- **L (Above Place again)**: The tool retracts upward after placing, returning to a safe clearance position

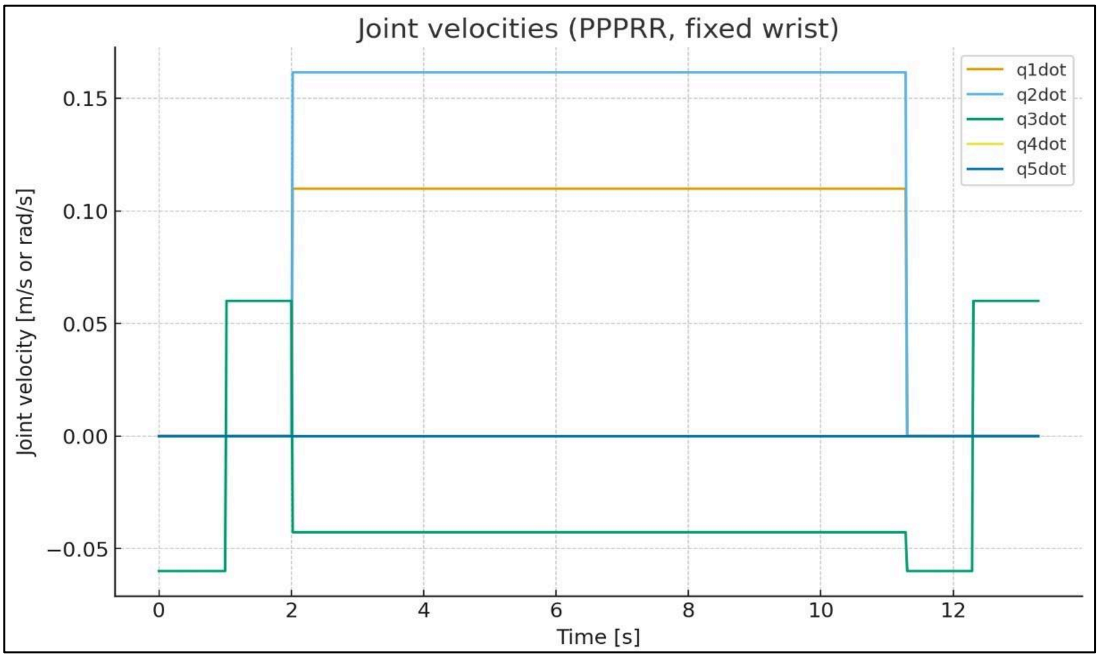

Figure 13 – Joint velocities of robot arm

3.7. ARTE simulation of assembly process

The ARTE simulation of the assembly process doesn't use the inbuilt animation functions or the tool attachments. The inbuilt animate function caused the robot to move immediately to the new position instead of a gradual motion.

The tool attachments and related functions weren't used as when a tool was attached to the robot arm, the pieces disappeared. Instead, after the arm moves to the part location, the graphics are enabled for a piece in that location.

4. Conclusion

The group was able to successfully design and simulate the model as per the requirement of the project. Several iterations were to be made to the design and some of the tasks had to be reassessed upon comparing the manual calculations (Section A) and simulation (Section B).

The group was unable to identify the reason as to why the Transformation matrix did not match between the manual calculation and ARTE; however, it was assumed that this may be an underlying difference in how ARTE interprets the parameters fed into the toolbox. The team found it to work in parallel to complete their tasks due to the sequential workflow of the project, it was important that each member communicate and share their updates with the rest of group to make steady progress. The project provided an umbrella understanding on the process of designing and simulating a manipulator for various tasks such as assembling a robot.

5. APPENDIX

A. Fair Contributions Sheet

Group 2
Enter the average fair contribution scores here:

Student Name	Student Number	Mark	Signature
Ashar Masood	7880777	10	Ashar M
Emily Arneodo	9991918	10	Emily A.
Jason Hurnall	6784707	10	Jason H.
Mathias Malandu	7162297	10	Mathias M.
Muhammad Ansari	7731218	10	Muhammad A.
Vidhukrishnan Pillai	9791255	10	Vidhukrishnan R
Yasan Gunathilaka	7549246	10	Yasan G.

B. Summary of scripts using in MATLAB

File path used:
/Users/ysemila07/Desktop/Yr 4 Sem 2/ECTE471/ARTE-master/robots/Project/robot_arm

Script	Section Applied	Description
jacobian_tool_config.m	Section A: 2.4	Calculates how the joint motion is used to relate to the end tools speed, direction and orientation.
parameters.m	Section B: 3.1	Configure robot arm's DOF, joint types, DH parameters, joint limits and other configurations.
import_links.m	Section B: 3.1	Import CAD files into ARTE and transform links to reference systems in moving frames. Convert dimensions millimetre to metre.
run_robot.m	Section B: 3.1	Load and simulate robot using the 'teach' function.
kinematics.m	Section B: 3.2	Derives the robot's full kinematic model by calculating transformation matrices from DH parameters.
jacobian.m	Section B: 3.3	Calculates and analyses the Jacobian matrix at pick/deposit positions and checks for singularities.
robot_tooltip.m	Section B: 3.4	Calculate the tool tip positions and orientations of the robot and visualise them.
smooth_transition.m	Section B: 3.5	Determining smooth tooltip trajectory between the nut feeder and the bolt.
trajectory_plot.m	Section B: 3.6	Draws the tool trajectory of M3 screw from pick to place and gives the graph of joint velocities.
assembly_animate.m	Section B: 3.7	Simulate assembly process of robot arm graphically using ARTE.